

Warping aerial photographs to orthomaps using thin plate splines

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Using an image rectification application arising in the field of forest management, we demonstrate in this paper the practical feasibility of applying thin plate spline (TPS) techniques to real image warping problems. The use of TPS-based warping in large problems can be limited by two factors: numerical instability in the calculation of TPS coefficients, and the intensive computation involved in evaluating TPS functions. Both drawbacks can be overcome by taking advantage of some recent advances in the numerical analysis of radial basis functions. Here we relate our experience in applying some of this work to realize successful TPS warping of large forestry images, and some graphical examples are given. Methods for automated control point selection and editing are also presented, and a cross-correlation technique for evaluating the effectiveness of the warps is described. This experience has guided our development of an effective and efficient software package for control point selection and TPS warping of digital images.

1. Introduction

Forest management applications in British Columbia, Canada, make extensive use of orthomaps, which are mosaics of rectified aerial photographs. These provide a highly useful tool for navigation, forest cover assessment, and other activities where an accurate and texturally descriptive land reference basis is required. Orthomaps, which typically have a pixel resolution of 1 m on the ground, are produced by a complex and costly process involving acquisition of aerial photographs and ground control data, aerial triangulation, and sophisticated processing of the raw images in association with a digital elevation model. Consequently, new versions of the full orthomaps are usually generated only every few years.

Of course, during these time intervals, the characteristics of the region will change due to both natural (e.g., wildfires, landslides, river course changes) and anthropogenic (e.g., road-building, logging) causes. Clearly, to satisfy orthomap-based forest management applications, it is essential that these maps be as up-to-date as possible. It is desirable to achieve this aim efficiently by obtaining recently acquired aerial photographs of areas known to have changed, rectifying these images, and updating the corresponding areas of the full orthomaps. Any successful realization of this approach, however, must overcome the problem of the geometric distortions that arise as a result

of the finite height of the aerial camera and the relief of the terrain being imaged. The combination of these effects can result in highly nonlinear, and locally variable, distortions in the images.

Warping techniques have been proposed as a means of reversing these distortions to a large extent, thus allowing the warped (rectified) image to replace the corresponding area of the orthomap. In general, warping involves three stages: (1) the definition of two sets of control points, corresponding to the same features in the two images (i.e., in the orthomap and the aerial photograph), (2) the selection and generation of coefficients for a pair of mathematical warping functions, and (3) the evaluation of these functions for all the pixel positions in the image to be warped. The complexity of these distortions requires the use of mathematically sophisticated warping functions in order to achieve acceptable results. In this connection, 2-D polynomials are generally found to be unsuitable, since for low degrees they do not provide sufficient “surface-fitting power”, and at higher degrees they can introduce oscillations, particularly in the edge regions of the image.

In our experience, the most suitable functions for warping applications characterized by nonlinear and local distortions are thin plate splines (TPSs). These are normally used as interpolating functions, and consequently ensure an exact correspondence between the control points in the two images. This is in contrast to least-squares techniques (involving, for example, polynomials) which provide an approximate fit at the control points. TPSs are locally sensitive functions that produce minimal oscillations at points away from the control points; thus, large local distortions can be accommodated without undesirable effects being propagated throughout the image. Indeed, TPSs are the “smoothest” functions which can provide an exact fit to sets of control points. For these reasons, in our company (BCS) they have become the preferred choice for warping applications, particularly for side-scan sonar image processing [1,2].

Until fairly recently, there have been two drawbacks in applying TPSs to practical image warping problems. The first is that numerical difficulties in computing the coefficients for the model can arise, and the second is that substantial computational times are required to evaluate TPSs for large numbers of image pixels. As described below, recent progress in numerical analysis has allowed both problems to be overcome.

For any warping scheme to be realized effectively, a sufficient number of control point pairs must be provided. Specification of control points is often done manually, although it is a laborious and error-prone process. We have therefore sought methods that would improve the speed of control point specification and validation. In this paper an effective technique is described for automated control point specification; this procedure is based on cross-correlation techniques applied to define similarity transforms in successive subdivisions of the images. Of course, these cross-correlation-based techniques are most effective in situations where the two images have a sufficiently high degree of similarity. This has indeed been the case in applications where we have

successfully applied these techniques; for example, forestry map updating¹ and mine countermeasures change detection.²

An interactive software system for control point editing is also described. This system allows semi-automated control point specification, and input of pre-selected control points; it also provides techniques for control point validation and editing.

Together, all these developments provide a suite of effective and efficient techniques for control point selection and TPS warping of digital images.

2. Gridding and TPS functions

Given a set of distinct control points $\{(x_i, y_i), i = 1, 2, \dots, n\}$ that are not all collinear, with associated function values $\{z_i\}$, it is often desirable to estimate corresponding function values at the node points of a large rectangular grid. While many approaches to this gridding problem are feasible, here we seek the smoothest possible real-valued function F satisfying

$$F(x_i, y_i) = z_i \quad \text{for } i = 1, 2, \dots, n, \quad (1)$$

where our measure of smoothness is the integral

$$I(F) = \iint_{\mathbb{R}^2} \left(\frac{\partial^2 F}{\partial x^2} \right)^2 + 2 \left(\frac{\partial^2 F}{\partial x \partial y} \right)^2 + \left(\frac{\partial^2 F}{\partial y^2} \right)^2 dx dy. \quad (2)$$

In [3] it is proved that the variational problem of minimizing (2) subject to the interpolation conditions (1) is solved uniquely by a TPS of the form

$$F(x, y) = a_0 + a_1 x + a_2 y + \sum_{i=1}^n \lambda_i r_i^2 \log r_i, \quad (3)$$

where:

- a_0, a_1 and a_2 are the coefficients of the planar term of the spline,
- λ_i is the coefficient for the i th spline term, and
- $r_i^2 = (x - x_i)^2 + (y - y_i)^2$.

In addition, the integral I in (2) is finite if and only if the λ_i 's have the property that

$$\sum_{i=1}^n \lambda_i = 0, \quad \sum_{i=1}^n \lambda_i x_i = 0, \quad \text{and} \quad \sum_{i=1}^n \lambda_i y_i = 0. \quad (4)$$

¹ Here, the overall characteristics of the images are very similar, with changes (e.g., road building, logging) in localized regions comprising a relatively small proportion of the images to be matched.

² This application involves change detection in side-scan sonar images of the same seabed region, where the changes are very small in size (e.g., a group of 2–3 pixels) but of crucial importance.

An alternative “engineering” motivation for the use of TPSs as smooth interpolation functions is provided in [4], where the small deflection equation for a thin metal plate of infinite extent is considered. If deflections z_i are specified at each of the control points (x_i, y_i) , the plate will require point loads at each control point in order to achieve the desired deflections. Again, it turns out that the mathematical model for this problem is a TPS of the form (3) whose coefficients determine the point loads. In addition, if the plate is required to “flatten out” away from the point loads, this is achieved by imposing equilibrium conditions on the plate so that the sum of the loads and their moments is zero (i.e., the conditions (4) pertain).

Thus, a TPS has the desirable properties that it asymptotically approaches a global planar model outside the region of the control points, and the effects at each control point are predominantly local, i.e., largely confined to an immediate region around each (x_i, y_i) .

3. Application of TPS gridding to image warping

Image warping is a particularly appropriate application of TPS gridding. Here, n pairs of control points are defined that link corresponding features in a *reference* image and a *new* image (sometimes called a *source* or *sensed* image). In the forestry application described below, the reference image is an orthomap, and the new image is an aerial photograph.

When applying TPSs to warping applications, the problem can be formulated so that, for all corresponding pairs of control points, the deviations in x and those in y are separately modeled as two TPS functions of x and y . Thus, the TPSs provide a pair of distortion surfaces that together map all the control points of one image onto the control points of the other image, and thereby provide a sensible means of mapping all the remaining pixels of one image onto the other. In addition, the direction of the warp must be established. There are two possible choices for the direction of the warp, namely:

- *Forward warping*: The control points used to define both TPSs are those from the new unwarped image, and the resulting spline functions give the deviations that would have to be applied to each pixel of the new unwarped image to map it to its corresponding position in the new warped image. The forward warping approach can be thought of as answering the question, “For each pixel value in the new unwarped image, where in the new warped image should it be put?” In practice, this method has the disadvantages that some pixel values in the new warped image can remain unspecified, and also that more than one pixel value in the new unwarped image may be assigned to the same pixel location in the new warped image.
- *Reverse warping*: The control points used to define both TPSs are those from the reference image, and the resulting spline functions provide the deviations to apply so as to identify the pixel in the new unwarped image which is to be used to provide the value for the new warped image. The reverse warping approach can be thought

of as answering the question, “For each pixel in the new warped image, from where in the new unwarped image should its value be taken?” This method is free from the above disadvantages of forward warping, although it does not always produce superior results. However, reverse warping was used in all the forestry warping experiments described below.

4. TPS coefficient generation

Let $\{(x_i, y_i), i = 1, 2, \dots, n\}$ be a set of n control points (distinct and not all collinear), each with associated z_i , for which a TPS function F satisfying the interpolation conditions (1) is sought. A unique TPS with $(n + 3)$ coefficients exists in the form

$$z_i = F(x_i, y_i) = a_0 + a_1x_i + a_2y_i + \sum_{j=1}^n \lambda_j r_{ij}^2 \log r_{ij}, \quad (5)$$

where $r_{ij}^2 = (x_i - x_j)^2 + (y_i - y_j)^2$, and the conditions (4) also apply. In the warping application, these z_i 's are the differences in the x coordinates, and also in the y coordinates, of corresponding control points from the two images.

The a_i 's and λ_j 's in (5) can be computed from the matrix equation $\mathbf{H}\mathbf{v} = \mathbf{b}$, where

$$\mathbf{H} = \begin{bmatrix} 0 & r_{21}^2 \log r_{21} & \dots & r_{n1}^2 \log r_{n1} & 1 & x_1 & y_1 \\ r_{12}^2 \log r_{12} & 0 & \dots & r_{n2}^2 \log r_{n2} & 1 & x_2 & y_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ r_{1n}^2 \log r_{1n} & r_{2n}^2 \log r_{2n} & \dots & 0 & 1 & x_n & y_n \\ 1 & 1 & \dots & 1 & 0 & 0 & 0 \\ x_1 & x_2 & \dots & x_n & 0 & 0 & 0 \\ y_1 & y_2 & \dots & y_n & 0 & 0 & 0 \end{bmatrix},$$

$$\mathbf{v}^T = (\lambda_1, \lambda_2, \dots, \lambda_n, a_0, a_1, a_2) \quad \text{and} \quad \mathbf{b}^T = [z_1, z_2, \dots, z_n, 0, 0, 0].$$

This system of equations can be extremely ill-conditioned, particularly for larger numbers of control points, and application of a numerically stable algorithm is essential when computing TPS coefficients. A review of available TPS algorithms is given in [6], where direct (i.e., finite) matrix methods are advocated for smaller values of n , and iterative algorithms are recommended for larger problems (say, when n exceeds 1000). In our forestry applications n has never exceeded a few hundred control points,³ and so we have employed a direct method to construct an orthogonal matrix (which is a product of $(3n - 6)$ Givens rotations) that reduces the interpolation problem to

³ We have occasionally used this coefficient generation software on much larger problems, but it is not efficient then when compared to iterative methods.

a square positive definite system of $(n - 3)$ equations. The resulting system is then solved to eventually provide stable computed values for all $(n + 3)$ TPS coefficients.

5. TPS evaluation on large grids

Direct evaluation of a TPS via the functional form (3) can be prohibitively time-consuming on a large grid because each node point requires the evaluation of n distinct terms involving the logarithm function. In image warping problems, the grids can easily be of size 2000×2000 points (pixels) or larger, and involve hundreds of control points. The fast TPS evaluation method employed in our warping software package, which is based on a tabulation scheme given in [5], replaces the direct evaluation form (3) with an approximation strategy that yields estimates accurate to within a user-specified tolerance. *The use of this approximation technique can result in evaluation times that are more than 1000 times faster than those for the direct TPS form (3), with minimal effects on the accuracy of the gridded values.* Table 1 gives a comparison of evaluation times for the fast and the direct TPS procedures for various grid sizes and numbers of control points. Notice that the last line of this table provides an example where use of the fast evaluation procedure reduces the evaluation time from nearly 5 hours to less than 14 seconds.

Table 1
Comparison of evaluation times for fast (approximate) and direct TPS gridding algorithms, on a Pentium 133 MHz computer with 64 Mbytes of RAM.

Number of control points	Times (sec):		Ratio of times
	Fast	Direct	
Grid size 300×300			
25	0.46	23.6	51
50	0.80	47.3	59
100	1.38	93.9	68
200	2.67	186.7	70
400	5.21	374.8	72
Grid size 1000×1000			
25	1.90	271	143
50	2.32	541	233
100	3.16	1082	342
200	4.86	2164	445
400	8.24	4328	525
Grid size 2000×2000			
25	6.44	1082	168
50	6.92	2164	313
100	7.81	4328	554
200	9.76	8656	887
400	13.31	17312	1301

6. Warping of aerial photographs to orthomaps

In this section we provide a summary of some recent work at BCS that demonstrated the practical feasibility of applying thin plate spline techniques to real image warping problems arising in forest management. The TPS warping technology outlined above was applied to six separate image pairs, obtained in regions of high relief in British Columbia. The aerial photograph images were pre-processed as required to give a new image of approximately the same scale (i.e., resolution) as that of the orthomap, and of a size to span a region slightly larger than that of the reference region in the orthomap. All six image pairs were compiled from aerial survey data that is accurate to within 10 m on the ground (latitude and longitude), and to within 5 m in altitude.

6.1. Control point selection and an example TPS warp

In the examples described here, control points were selected manually from the images. In all cases, we attempted to obtain a reasonably uniform spatial distribution of the points, and to only select those points where an unambiguous and spatially well-defined correspondence between each image pair could be determined. To minimize the effects of perspective and aspect, terrain features or the bases of trees were selected whenever possible. Under some conditions, as in densely forested areas, this was not possible, and the tops of trees were used instead. The sizes and numbers of control point pairs chosen for the six experiments are given in table 2.

An example of the performance of TPS warping applied to image 1 is given in figure 1. This figure illustrates the different scales of the reference and new image, and clearly demonstrates the effectiveness of the warping. In particular, the strong distortions in the region in the top right of the new image are seen to be effectively corrected in the warped image.

6.2. Cross-correlation analysis of TPS warping results

In evaluating the consistency and effectiveness of the warping, one approach is to estimate how closely regions in one image are mapped to corresponding regions in the other. This may be accomplished by multiple cross-correlation analysis, where the image is divided into a number of sub-regions, and the overall degree of feature

Table 2

Image#	Reference image size	New image size	Number of control point pairs
1	1850 × 1850	1920 × 1920	167
2	1850 × 1850	1920 × 1920	140
3	1850 × 1850	1920 × 1920	152
4	2100 × 2100	2500 × 2500	185
5	1850 × 1850	2200 × 2200	156
6	1850 × 1850	2200 × 2200	138

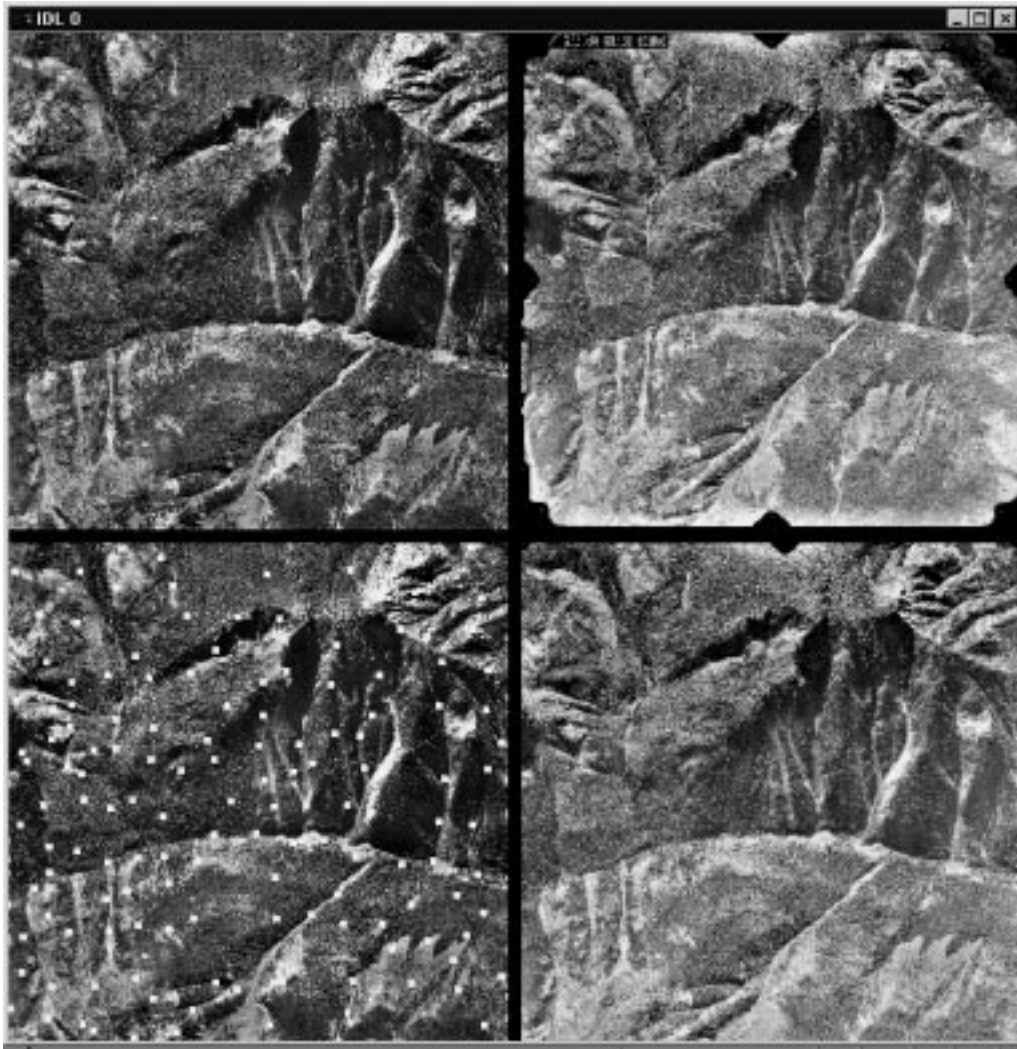


Figure 1. TPS warping of aerial photograph onto orthomap for image 1. Top left: orthomap (reference image); top right: aerial photograph (new image to be warped); bottom left: reference image with positions of control points indicated; bottom right: TPS-warped new image.

correspondence is estimated within each separate region. The results of this process can then be analyzed statistically to express the shifts in the features (which ideally should be zero for identical images/perfect warps) in terms of means, or percentiles.

Shifts may be estimated using cross-correlation techniques. The cross-correlation function P for two images, A and B, is a 2-D function defined as

$$P(k, l) = \frac{\sum_i \sum_j (A_{ij} - \bar{A})(B_{i+k, j+l} - \bar{B})}{[\sum_i \sum_j (A_{ij} - \bar{A})^2]^{1/2} [\sum_i \sum_j (B_{ij} - \bar{B})^2]^{1/2}},$$

Table 3

Image#	Mean correlation	Mean shifts			Percentile abs shifts		
		x	y	abs	50%	75%	90%
1	0.563	-0.14	-0.24	1.80	1.00	2.24	4.47
2	0.433	-0.22	-0.17	2.20	1.41	2.24	4.12
3	0.546	-0.15	-0.04	1.25	1.00	1.41	2.24
4	0.431	-0.18	-0.10	2.28	1.41	2.83	5.00
5	0.563	-0.14	-0.24	1.80	1.00	2.24	4.47
6	0.471	-0.24	0.69	1.95	1.41	2.24	4.12

where:

- A_{ij} and B_{ij} are the intensity values at the (i, j) th position of images A and B, respectively,
- $\bar{A}$ and $\bar{B}$ are the mean intensity values of images A and B, respectively,
- k and l are the (integer) shifts in the x and y components of A and B, respectively, at which the cross-correlation value is evaluated.

For this analysis, 144 evenly spaced sub-images, each of size 128×128 pixels, were derived from each image pair. The full 2-D cross-correlation function was then computed for each sub-image pair, using fast Fourier transform techniques for efficiency, and the maximum value in this function was identified. (This generated the circular cross-correlation function, but for the small shifts usually observed in the sub-images being correlated, any edge effects are minor.) The position of this maximum value corresponds to the shift in x and y that yields the closest match between the sub-images.

This analysis was applied to the six image pairs used in this study. For each pair, the average shifts in x and y for the 144 sub-images were computed, along with the average absolute shifts. In addition, the percentiles of the distributions of the absolute shifts were computed, in order to determine threshold shift values below which observed values would be expected to fall a certain percentage of the time.

The results of this cross-correlation feature shift analysis are given in table 3. All shifts are expressed as meters.

The results for these images confirm that features in the TPS-warped new image correspond with those in the reference image to within a very few pixels. From this cross-correlation analysis, it may be concluded that, for images of the type encountered in the present application, *the TPS-warped features were corrected with great effectiveness, being accurate 50% of the time to within 1.5 m, and 90% of the time to within 5 m.*

6.3. Polynomial warping example

To provide a comparison with TPS warping, polynomial warping was applied to image 1 by fitting least-squares polynomials of degrees 1 to 11 at the same control

points as used for TPS warping. The 2-D polynomials of degree k used were defined as follows:

$$P(x, y) = \sum_{i=0}^k \sum_{j=0}^k c_{ij} x^i y^j.$$

Hence, a degree k polynomial has $(k + 1)^2$ coefficients, and contains a term in $x^k y^k$.

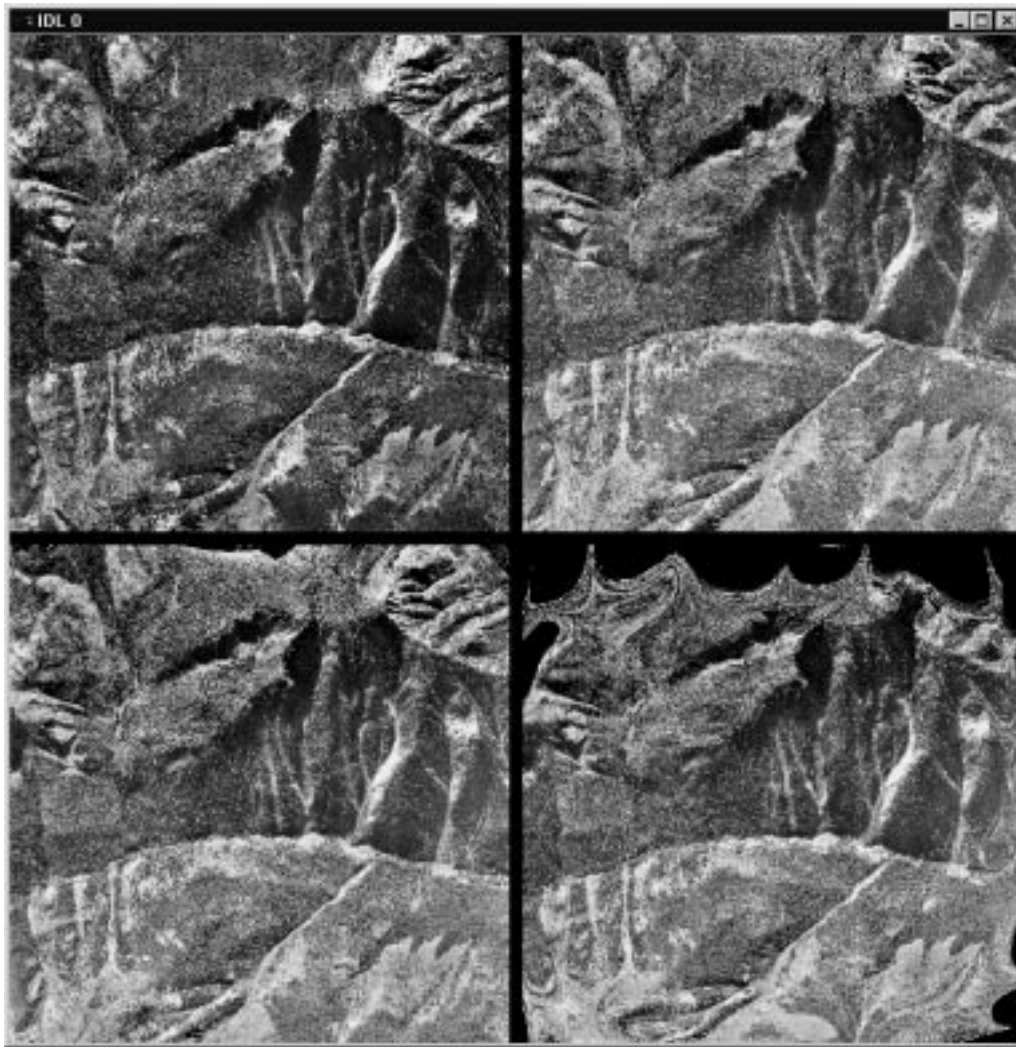


Figure 2. Polynomial warping of aerial photograph onto orthomap for image 1. Top left: orthomap (reference image); top right: polynomial warp (degree 2) of aerial photograph; bottom left: polynomial warp (degree 7) of aerial photograph; bottom right: polynomial warp (degree 11) of aerial photograph.

Special-purpose routines, involving double precision, appropriate matrix scaling, and singular value decomposition, were written to compute accurate values for the coefficients for the polynomial warping functions, and double precision arithmetic was also used to evaluate the polynomials themselves.

As illustrated in figure 2, two types of limitations were observed for polynomial warps. The first type, which was observed when the degree was low (say, one or two), was due to the fact that a low degree polynomial warping function is not complex enough to model the image distortions adequately. This is illustrated in figure 2 (upper right image), where the degree-two warp produced a result which did not effectively correct for the strongly nonlinear distortions in the top-right corner. The second type, which became more pronounced as the degree increased, was the introduction of distortions, particularly near the edges of the image. This is shown in figure 2 (lower right image), where the degree-eleven warp gave a severely distorted result for edge regions of the image. This was due to the combined effects of the relative lack of control points near the edges and the oscillatory nature of polynomial functions. Even for optimal degree (seven), some evidence of distortion was evident in the edge region of the images (figure 2, along top of lower left image).

These and other results confirm in our opinion the superiority of TPSs over polynomials in their effectiveness as warping functions.

7. Improvements in control point specification

A sufficient number of control points must be provided for image warping to be effectively realized. As in the warping experiments described above, control point selection is often done manually, and can be a time-consuming and error-prone process. It is therefore highly desirable to devise approaches that automate the selection process as far as possible and which allow the effective editing of control points. This section describes some techniques recently developed by BCS to achieve these aims.

7.1. Fully automated control point specification

In this approach, control point pairs from a reference image and a new image are generated by means of correlation/subdivision techniques. The basic idea is to first estimate a linear similarity transform relating the reference image and new image, where this transform is characterized by four parameters, i.e., rotation about a point, magnification, and shifts in x and y . The parameters of this transform can then be used as initial values for estimating parameters for four new similarity transforms, one for each of the four quadrants of the reference image. Each of these four transforms is then used to estimate new transform parameters for each of the four sub-quadrants (sixteen in all), and so on, until transform parameters for each of a sufficiently detailed number of regions have been defined. For each of these regions, the transform can be used to define which point in the new image maps best into the point in the center of the region of the reference image. Thus the parameters of an initial global transform are used

Table 4

Selection scheme	Mean correlation	Mean shifts			Percentile abs shifts		
		x	y	abs	50%	75%	90%
Manual	0.563	-0.14	-0.24	1.80	1.00	2.24	4.47
Automatic	0.553	-0.76	-0.26	2.73	1.41	2.23	6.08

to generate successive transforms which are progressively more local in nature, and which ultimately yield estimates for control point pairs which are distributed uniformly over the images.

In realizing this approach in practice, it was essential, for computational efficiency, that the image regions to be matched were reduced to a manageable size (say, 128×128) from their original sizes before matching. Image matching was done using cross-correlation techniques. Depending on the type of matching to be done, this could involve either fast Fourier transform-based methods (when the cross-correlation at all possible shifts was required) or spatial domain summation (when a single correlation value for fixed shifts was required). At each stage in the subdivision process, the initial estimates of the transform parameters were refined, by first estimating the shifts in x and y , and then sequentially estimating the peak in the correlation function for each parameter, using quadratic interpolation. In this connection, smoothing of the images before cross-correlating was found to be useful in correctly identifying the position of the peak. Typically, subdivision was continued until transform parameters for 256 separate regions (thereby defining 256 control point pairs) were obtained. Only those control point pairs from regions where the correlations exceeded a specified threshold were output.

An example of the performance of the automated control point specification algorithm compared to manual selection is shown in table 4. The algorithm was applied to generate control point pairs for image 1, and the pairs with correlations exceeding the threshold of 0.7 (i.e., 202 pairs out of 256) were used to perform a TPS warp on the image. The cross-correlation analysis scheme was applied to the resulting warped image, and the results of this are compared in the table below with those obtained for TPS warping using the manually selected 167 pairs. These results show that the (unedited) automatically generated control points produced nearly as high a degree of correspondence between the warped image and the reference image as using the manually selected control points.

Thus, the automated control point specification approach can provide a highly effective means of generating reliable and accurate control point pairs between the two images. The control points generated by this process may then be examined and edited by the methods described in the next section.

7.2. Control point editing and semi-automated specification

This section describes the control point editor software system developed by BCS. This system provides a graphical user interface (GUI) for the specification, input and

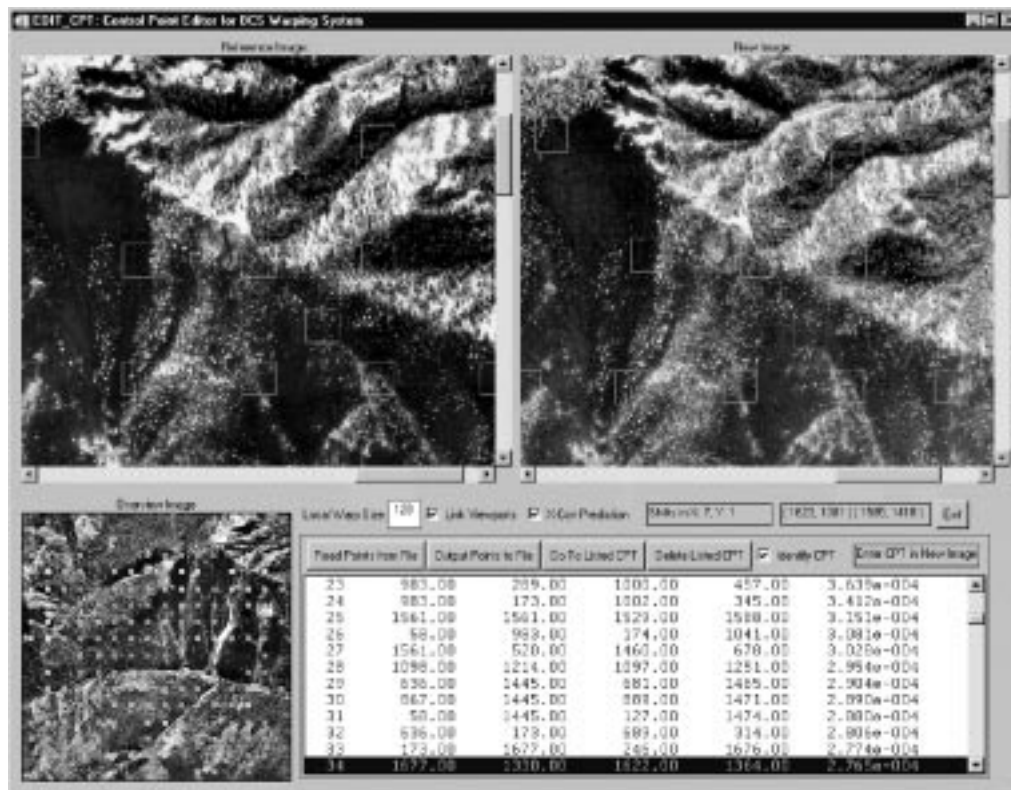


Figure 3. Graphical user interface for control point specification and editing.

editing of control point pairs in the reference and the new images, with the aid of approximate or accurate prediction of location. It also allows the input of a previously generated set of control point pairs, the assessment of the validity of these points, and the provision for editing them. The output of this component is a set of verified control points, which may then be used in performing a TPS warp of the new image with respect to the reference image.

Some of the main functional components of this GUI, which is illustrated in figure 3, are:

- side-by-side display of the reference and new images, with boxes outlining the control point locations;
- display of a reduced version of the entire reference image in an overview window, with the locations of all control points indicated;
- an option to link the region displayed for the new image to the region currently displayed for the reference image;
- manual specification of control point pairs;

- dynamic *approximate* control point prediction in the new image, based on the current TPS model;
- dynamic *accurate* control point prediction in the new image, based on the current TPS model and cross-correlation methods, with display of the predicted shifts in x and y ;
- an option to perform and display the results of a local warp using the current TPS model;
- user-controlled acceptance of control point pairs;
- a list of the current control point pairs and the magnitudes of the TPS coefficients corresponding to each pair;
- an option to select a control point on the list and indicate its location on the images;
- an option to select a control point on an image and identify it on the list;
- an option to delete a selected control point pair from the list;
- a means of preventing the occurrence of two control points within a small specified distance of each other;
- an option to read in a set of control point pairs from a file and initiate the control point specification/editing session; and
- an option to output the current list of control points to a file.

Some of the major features of this interface are now described in more detail.

7.2.1. Reference and New Image

These windows indicate the locations of existing control points (shown as green outline boxes) and allow the specification and editing of control points. As the cursor is moved within the Reference Image window, a red outline box indicates its position within this window. Simultaneously, a dynamic “ghost” outline box in the New Image window indicates the predicted position of the point in the new image corresponding to the cursor position in the reference image. This prediction may be either *approximate* or *accurate*, as described in section 7.2.3.

Pressing the left mouse button with the cursor in the Reference Image window selects a control point for the reference image there and displays a green outline box at this location. Once this point is selected, a corresponding control point in the New Image window must then be manually specified; when this is specified, the cycle may be repeated. An efficient alternative to manual specification in both windows can be performed if the predicted position in the New Image window is acceptable; in this case, pressing the right mouse button results in automatic acceptance of the outlined points in the reference and new images. Each time a control point pair is added (or deleted), the TPS coefficients are updated.

A particularly useful feature provided in the Reference Image window is the ability to preview the local warp corresponding to the current TPS model. When the center mouse button is pressed, a TPS warp of the new image is performed for a local

region of the reference image centered on the cursor location, and this is overlaid on this region of the reference image. Releasing the button restores the original region of the reference image. This real-time warping feature enables the immediate assessment of the quality of the current warping for a particular region, and provides an indication of whether more control points may be required.

7.2.2. *Overview Image*

This window contains an overview of the entire reference image, sub-sampled so as to fit into the space available. The locations of the current control points are indicated on this image as small squares, allowing convenient assessment of the distribution and adequacy of sampling of the control points.

7.2.3. *X-Corr Prediction*

This toggle button selects the type of prediction used to automatically define the position of the cursor in the New Image window. If this option is not set, then the current TPS warping function is evaluated for the single point at the current cursor position, and this result is used as the *approximate* prediction. If it is set, then the current TPS warping function for a surrounding region is used in conjunction with an iterative cross-correlation technique to give an *accurate* prediction of the location of the control point. In addition, when this option is set, the predicted shifts in X and Y (i.e., the accurate prediction compared to the approximate prediction) are displayed in an associated text window.

These predicted shifts provide a very useful indication of the adequacy of the current control points in defining the TPS warp for the region surrounding the cursor. When the shifts are low, the quality of the warp will be satisfactory, but when they are high, then it will be necessary to specify additional control points, or modify existing control points as appropriate. Because of its use of cross-correlation techniques, the X-Corr Prediction option can be used to specify control points even when no distinct features are present near the cursor. This makes it very useful for regions of the images where the information is represented more in terms of textures rather than distinct features.

The X-Corr Prediction option thus provides for both semi-automated selection of control point pairs (when used with the right mouse button) and assessment of the correctness of previous control point selection/prediction (for example, when a file of pre-estimated control points is used). This feature provides an extremely powerful and convenient mechanism for control point specification and assessment.

7.2.4. *List of control points*

This list contains information about the locations and corresponding TPS coefficients of the current control points. For each point, its position in the list, the X and Y values of the point from the reference image, and the X and Y values for the point from the new image are exhibited. The final field contains the magnitude of the TPS

coefficients for that control point (i.e., $\sqrt{(\lambda_x)_i^2 + (\lambda_y)_i^2}$, where $(\lambda_x)_i$ and $(\lambda_y)_i$ are the TPS coefficients corresponding to the i th control point for the TPS warping functions in the x and y directions, respectively).

The control points are listed in decreasing order of magnitude of the TPS coefficients. This ordering is based on the observation that the magnitude of each TPS coefficient is related to the size of the local distortion. Consequently, this ordering provides a convenient way of detecting those control point pairs with the largest local deviations, i.e., those pairs which are most likely to have been incorrectly matched. The ordering allows for efficient examination and editing of a set of previously estimated control point pairs, in which some matching errors may be present, since these errors will usually result in larger magnitudes for the TPS coefficients.

8. Conclusions

The results summarized in this paper demonstrate the practical feasibility of applying thin plate spline warping techniques to image rectification problems. Specifically, we have shown via some forest management examples that TPSs can provide warped updates to images that are as accurate as the data warrant. This satisfactory state of affairs has been made possible by recent progress in numerical analysis. In addition, the process of specifying control points to be used in the warping has been greatly speeded up through the development of an automatic control point specification procedure (for when the images are sufficiently similar) and a versatile control point editing system. Together, these advances now allow the benefits of TPS warping to be efficiently applied to practical image warping problems.

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